

SPLITEASY

Comprehensive Project Architecture & Development Report

A Technical Documentation Project

Project Information

Field	Details
Project Name	SplitEasy
Repository	spliteasy
Project Type	Full-Stack Web Application
Domain	Expense Management & Financial Technology
Version	0.1.0
Status	Phase 5 Complete, Phase 6 In Progress
Date	April 2026

Student Information

Name	UID
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Academic Details

- **Degree Program:** Bachelor of Science (B.Sc.)
- **Field of Study:** Computer Science and Engineering
- **Specialization:** Data Science
- **Department:** Department of Computer Science and Engineering
- **Submission Date:** April 2026

Executive Summary

This comprehensive technical report documents the complete architecture, implementation, and analysis of **SplitEasy**, a modern full-stack expense-splitting web application. The project demonstrates advanced knowledge of:

- **Full-Stack Development:** Next.js, React, TypeScript, MongoDB
- **Software Architecture:** Component design, API structure, data modeling
- **Software Engineering:** Type safety, security practices, scalability
- **UX/UI Design:** Responsive design, accessibility, mobile optimization
- **Data Science Applications:** Balance calculations, settlement algorithms, data analysis

Project Quality Assessment:

- Architecture: 8/10 
- Code Quality: 8/10 
- Features: 8/10 
- Mobile UX: 3/10  (Phase 6 focus)
- Overall: 6.0/10 (Solid foundation, needs mobile polish)

Report Structure

This report is organized into the following sections:

1. **Executive Summary** - High-level project overview
2. **Project Overview** - Purpose, scope, value proposition
3. **Tech Stack** - Complete technology inventory
4. **Architecture** - System design, structure, patterns
5. **Core Features** - Feature breakdown and workflows
6. **UI/UX Design** - Visual design, layout, components
7. **Mobile Responsiveness** - Current state and gaps
8. **Strengths** - What's working well (validated)
9. **Limitations & Issues** - Problems and challenges
10. **Recommendations** - Detailed roadmap (Phase 6-10)
11. **Quality Scorecard** - Comprehensive evaluation

Document Metadata

- **Document Type:** Technical Project Report
- **Length:** Comprehensive (40+ pages)
- **Purpose:** Technical documentation, code review, project planning
- **Audience:** Development team, academic reviewers, stakeholders
- **Format:** Markdown (convertible to PDF, DOCX, HTML)

Generated: April 25, 2026

Version: 1.0

Status: Complete & Ready for Review
